CONTINUOUS ASSESSMENT INDIVIDUAL MINI-PROJECT
COURSE: GENERATIVE AI (B.TECH CSE)

# NeuroTutor AI: AI–ML–DL Concept Tutor

An Adaptive Multi-Tier Educational Platform with Real Attention Extraction, Notes RAG Semantic Search, PyTorch GANs, and Responsible AI Grounding

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| **Roll Number:** | 1 |
| **Allocated Problem:** | P1 — AI–ML–DL Concept Tutor |
| **Degree & Department:** | B.Tech in Computer Science & Engineering |
| **Course Code & Title:** | Generative Artificial Intelligence |
| **Project Submission Folder:** | 1_Aarush_Parate/ |
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**Project Abstract:** NeuroTutor AI is an end-to-end Generative AI educational platform engineered to bridge the gap between high-level intuitive AI concepts and low-level mathematical implementations. Designed to cover all five compulsory course units, it features a 3-tier adaptive concept tutor (School, Undergraduate, Expert), a Prompt Engineering Lab (Zero-Shot, Few-Shot, Chain-of-Thought), local DistilBERT 72-head attention heatmaps, Sentence-BERT (all-MiniLM-L6-v2) FAISS semantic notes retrieval with strict uncertainty refusal, multimodal diagram visual parsing, hosted diffusion concept illustration, PyTorch adversarial GAN bimodal training, and a 93.3% accurate hallucination grounding verification suite.

# 1. Problem Statement and Primary Objectives

### Problem Statement (P1):

Undergraduate engineering students frequently struggle to master Artificial Intelligence, Machine Learning, and Deep Learning due to disjointed learning materials that either oversimplify concepts into imprecise analogies or overwhelm learners with complex equations divorced from operational code. Furthermore, standard public LLM chatbots hallucinate unsupported facts, lack verifiable grounding against curriculum notes, provide zero visibility into intermediate attention representations, and fail to provide hands-on neural network demonstrations.

### Primary Engineering Objectives:

1. **Multi-Tier Adaptive Tutoring:** Explain any AI/ML/DL concept across 3 distinct knowledge levels: School Level (everyday analogies), Undergraduate Level (algorithms & pseudocode), and Expert Level (formal research derivations).

2. **Prompt Engineering Analysis:** Compare Zero-Shot, Few-Shot, and Chain-of-Thought prompting side-by-side with dynamic softmax temperature simulation (0.0 to 1.5).

3. **Transformer Interpretability:** Extract real multi-head self-attention weight matrices from local Hugging Face DistilBERT models across 6 layers and 12 heads (72 subspaces).

4. **Grounded Academic Retrieval:** Ingest lecture materials (PDF, DOCX, TXT), compute 384-d dense embeddings with Sentence-BERT, and perform FAISS vector similarity search with uncertainty refusal fallback.

5. **Multimodal Studio & Speech:** Parse architectural schematics with BLIP/Gemini Vision, generate educational diffusion illustrations, and synthesize spoken voice summaries via gTTS.

6. **PyTorch Demonstrations:** Train an adversarial GAN on bimodal continuous distributions in real time and inspect backpropagation autograd gradient flows in an educational MLP.

7. **Responsible AI & Grounding Verification:** Evaluate claim grounding, defend against prompt injections, and evaluate performance on a labelled benchmark dataset.

# 2. Tools, Models, and Datasets

| System Component | Tool / Framework / Model | Execution Mode | Role & Purpose |
|---|---|---|---|
| Frontend UI | Streamlit 1.51 + Plotly + CSS | Local Web App | Modern dark-navy SaaS dashboard with interactive charts |
| Transformer Model | Hugging Face DistilBERT (6-L, 12-H) | Local PyTorch CPU | Real multi-head attention weight extraction |
| Embedding Model | Sentence-BERT (all-MiniLM-L6-v2) | Local Inference | 384-dimensional dense semantic vectors for notes RAG |
| Vector Index | FAISS CPU (IndexFlatIP) | Local In-Memory | Exact cosine similarity semantic passage retrieval |
| Generative LLM | Google Gemini 2.5 Flash / Groq | Cloud API + Offline | Multi-turn reasoning, prompt synthesis, and grounded QA |
| Multimodal Vision | Gemini Vision / BLIP Heuristics | API + Local Heuristic | Architectural schematic parsing and component identification |
| Concept Illustration | Hosted Diffusion API (Pollinations) | Hosted HTTP Inference | Prompt-enhanced educational diagram generation |
| Speech Synthesis | gTTS (Google Text-to-Speech) | Python Library | Natural speech audio narration with accent controls |
| Deep Learning Core | PyTorch 2.9 (CPU) | Local PyTorch Execution | Live bimodal GAN training & MLP autograd backprop |

## 3. Technical Approach and System Architecture

NeuroTutor AI is architected as a modular, layered educational platform. The system completely decouples the presentation layer (Streamlit and Plotly) from core backend computation (LLM reasoning, vector embeddings, PyTorch autograd, and safety filters).

### Architectural Workflow Pipeline:

1. **Input Layer & Security Filter:** User prompts and uploaded documents are scanned by SafetyGuardrails for prompt injection signatures and unauthorized instruction overrides before processing.

2. **Cognitive Level Conditioning:** User-selected level (School, UG, Expert) conditions prompt templates and vocabulary constraints.

3. **RAG Context Injection:** Ingested notes are extracted via DocumentProcessor, chunked into overlapping passages, embedded into 384-d vectors with Sentence-BERT, and indexed in FAISS. The top-k matching passages are retrieved with cosine similarity scores.

4. **Dual Engine Routing:** Queries route to Gemini 2.5 Flash / Groq when API keys are available, or to NeuroTutor's deterministic knowledge engine when operating offline.

5. **Grounding Verification:** Answers are verified by HallucinationChecker against retrieved passages, assigning Supported, Partially Supported, or Not Supported status.

6. **Visual & Auditory Augmentation:** Spoken audio is generated via AudioSummaryGenerator, while real DistilBERT attention weights and PyTorch GAN loss curves render interactively.

## 4. Key Implementation Details and Code Explanations

### A. Real Attention Extraction (Hugging Face DistilBERT):

Unlike mock heatmaps, NeuroTutor AI runs a local DistilBERT transformer with output_attentions=True. It extracts raw pairwise attention weights $A_{ij} = \text{softmax}(Q_i K_j^T / \sqrt{d_k})$ across 6 layers and 12 heads (72 subspaces):

```
# modules/attention_visualizer.py
inputs = tokenizer(text, return_tensors='pt', truncation=True, max_length=32)
with torch.no_grad():
    outputs = model(**inputs)
# outputs.attentions tuple: 6 layers, each shape [1, 12, seq_len, seq_len]
layer_np = outputs.attentions[layer_idx][0].cpu().numpy()
```

### B. Sentence-BERT & FAISS Vector Indexing (RAG):

```
# modules/rag_engine.py
embeddings = embedder.encode(texts, convert_to_numpy=True).astype(np.float32)
faiss.normalize_L2(embeddings)  # Normalize for exact cosine similarity
index = faiss.IndexFlatIP(384)  # Inner-Product FAISS CPU Index
index.add(embeddings)
```

### C. PyTorch Real-Time GAN Adversarial Training:

```
# modules/gan_explorer.py
loss_d = criterion(D(real), 1.0) + criterion(D(G(z).detach()), 0.0)
loss_d.backward(); opt_d.step()
loss_g = criterion(D(G(z)), 1.0)
loss_g.backward(); opt_g.step()
```

### Public GitHub Repository Link:

■ https://github.com/Aarush-Parate/neurotutor-ai

## 5. Application Visual Architecture & Layouts

NeuroTutor AI is styled using a modern SaaS dark-navy theme (#2A3140 primary background, #3C4959 surface cards, #728EA6 accents). Below is the visual map of the 11 integrated modules accessible via the sidebar navigation:

| Page Section | Key Features & Visual Components |
|---|---|
| 1. Dashboard | Executive overview, 3-tier learning level badges, 5-unit syllabus cards, and technology stack matrix. |
| 2. AI Concept Tutor | Conversational dialogue bubbles, sliding-window session memory, knowledge level dropdown, temperature slider, and inline audio playback. |
| 3. Prompt Engineering Lab | Side-by-side Zero-Shot, Few-Shot, and Chain-of-Thought columns, interactive temperature softmax bar chart with Shannon entropy, Gen vs Discrim decision boundaries, and 2D latent space scatter plot. |
| 4. Attention Explorer | Interactive Plotly heatmap of real DistilBERT attention weights, layer and head selectors, token entropy metric, and academic interpretability caveats. |
| 5. Knowledge Base & RAG | Multi-format note ingestion (PDF/DOCX/TXT), pre-packaged course notes loader, FAISS index statistics, grounded answer card with source citations, and GPT vs BERT comparison tab. |
| 6. Diagram Explainer | Visual diagram uploader, pre-packaged architecture schematics (Transformer, CNN, GAN), multimodal analysis breakdown, and transparent vision limitation disclosures. |
| 7. Concept Image Studio | AI concept presets, automated prompt expansion preview, hosted diffusion model generation, high-res preview, and PNG download button. |
| 8. Audio Learning | Text-to-Speech narration studio with Markdown/LaTeX cleaning, US/UK/Indian accent controls, audio playback widget, and MP3 download. |
| 9. GAN & PyTorch Lab | Real-time PyTorch GAN bimodal distribution training with interactive loss curves, plus a transparent 3-layer MLP autograd demo with L2 gradient norms. |
| 10. Responsible AI | Interactive grounding verifier, empirical evaluation dashboard with Plotly confusion matrix, adversarial prompt injection testing terminal, and licensing terms. |
| 11. About & Model Card | Formal Mitchell et al. Model Card, complete model transparency manifest table, and student submission details. |

## 6. Quantitative Results and Evaluation Tables

| Evaluation Dimension | Target Metric | Measured Performance | Academic Benchmark Status |
|---|---|---|---|
| Semantic Retrieval (RAG) | Top-3 Hit Rate on 10 Queries | **70.0%** | Exceeds baseline syllabus recall; 12.9 ms latency |
| Grounding Verification | 15-Case Labelled Accuracy | **93.33%** | 14/15 cases correctly identified |
| False Refusal Safety | Rejection on Supported Queries | **0.0%** | Zero false refusals on valid course facts |
| Attention Resolution | Hugging Face DistilBERT | **72 Subspaces** | 6 layers × 12 heads fully extracted |
| Prompt Technique Latency | Average Inference Time | **0.152s** (CoT) | Efficient execution under local caching |

## 7. Technical Limitations and Documented Failure Case

### Systemic Technical Limitations:

1. **Heuristic Grounding Boundaries:** Lexical token-overlap and heuristic verification methods can underestimate candidate answers that express valid facts using novel synonyms or non-verbatim paraphrases.

2. **Base Vision Coarseness:** Standard vision-captioning models identify broad structural shapes ('boxes and directional arrows') but cannot reliably parse microscopic subscripts or proprietary flowchart symbols without high-cost OCR vision backbones.

3. **Hardware & Memory Budgets:** To guarantee universal deployment on free CPU tiers (Hugging Face Spaces / Streamlit Cloud), diffusion image generation relies on hosted inference endpoints rather than local 12GB VRAM diffusion pipelines.

### Documented Real Failure Case Analysis (Mandatory Requirement):

| | |
|---|---|
| **Failure Case Identifier:** | FAIL-01 |
| **Failure Category:** | Heuristic Lexical Grounding Underestimation on Novel Paraphrases |
| **Input Question:** | *'What is mode collapse in GANs and how does Wasserstein GAN solve it?'* |
| **Expected Label:** | Partially Supported |
| **Observed Behavior:** | When an answer correctly identifies mode collapse from the context but accurately introduces outside knowledge regarding WGAN Earth Mover's distance, pure lexical token overlap scored 0.38, marginally crossing beneath the 0.40 threshold into 'Not Supported' on strict runs. |
| **Root Cause Analysis:** | Heuristic token overlap does not measure non-verbatim semantic synonymy or conceptual entailment as robustly as cross-encoder Natural Language Inference (NLI) models. |
| **Concrete Remediation:** | Integrate a lightweight Cross-Encoder NLI model (e.g. `cross-encoder/nli-deberta-v3-small`) to compute entailment, neutral, and contradiction logits directly over sentence-passage pairs. |

## 8. Ethical Considerations, Safety and Intellectual Property

**1. Adversarial Robustness & Prompt Injection:** Retrieved academic notes are treated as untrusted data inputs, defusing prompt injection payloads hidden inside student lecture notes. User inputs are screened against malicious system-override signatures.
**2. Student Data Privacy:** Uploaded PDFs and chat sessions reside strictly in ephemeral volatile RAM. No student work or proprietary course slides are saved to permanent disks or commercial training databases.
**3. Uncertainty Transparency:** When reference notes lack supporting facts, the system emits *'I'm not sure based on the available notes'* rather than fabricating plausible hallucinations.
**4. Open Source Licensing:** Built upon open-source foundations (DistilBERT: Apache 2.0, Sentence-BERT: Apache 2.0, FAISS: MIT, PyTorch: BSD-3). Core application released under the MIT License.

## 9. Conclusion

NeuroTutor AI successfully fulfills all academic requirements of the Continuous Assessment Mini-Project (P1) by seamlessly unifying all five syllabus units of Generative AI into a production-grade, highly engaging educational platform. Through its 3-tier cognitive explanations, real DistilBERT 72-head attention heatmaps, FAISS semantic notes retrieval, live PyTorch GAN training, and 93.3% accurate hallucination grounding verification, NeuroTutor AI establishes a rigorous, transparent benchmark for AI-assisted technical education.